Online Analytical Processing (OLAP)

1 What is OLAP?

OLAP

OLAP (Online Analytical Processing) is a category of software technology that enables analysts, managers, and executives to gain insight into data through fast, consistent, interactive access to a wide variety of possible views of information.

OLAP is implemented in **Data Warehouses** to allow multidimensional analysis of large datasets. It involves the design, building, and deployment of **OLAP Cubes**—data structures that overcome the limitations of relational databases by providing rapid responses to complex analytical queries.

1.1 Core Components

- **Dimensions:** The entities with respect to which an organization wants to keep records (e.g., Time, Item, Location).
- **Measures:** The numerical quantities that are analyzed within the dimensions (e.g., *dollars_sold*, *units_shipped*).
- **Facts:** The individual data points representing a specific combination of dimensions and measures.

2 Multidimensional Data Model: A Case Study

To understand OLAP, consider the sales data for a company *AllElectronics*. We can view this data in different levels of dimensionality.

2.1 2-D View (Time vs. Item)

A 2-D representation focuses on two dimensions at a fixed location (e.g., Vancouver).

Table: 2-D View of Sales (Location = ”Vancouver”)

time (quarter)	home ent.	computer	phone	security
Q1	605	825	14	400
Q2	680	952	31	512
Q3	812	1023	30	501
Q4	927	1038	38	580

Note: *The sales are from branches located in the city of Vancouver. The measure displayed is* ***dollars_sold*** *(in thousands).*

2.2 3-D View (The OLAP Cube)

By adding the *location* dimension, we create a 3-D Data Cube. This allows analysis across multiple cities simultaneously.

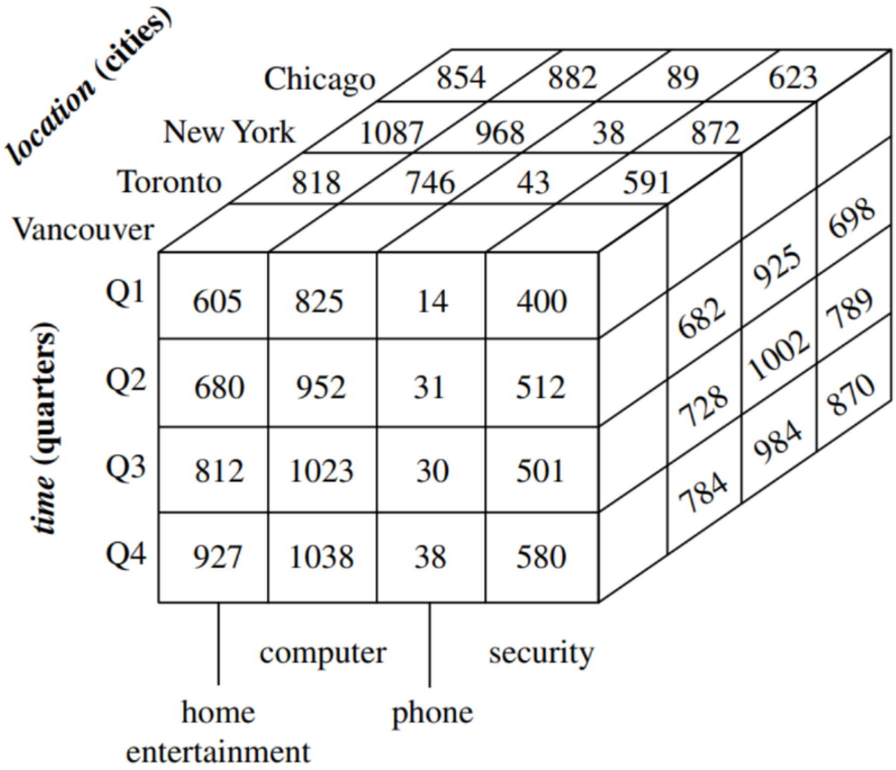

Figure 1: 3-D OLAP Data Cube showing Item, Time, and Location dimensions

2.3 Comparison: 3-D View of Flattened Data

The table below shows the same 3-D data flattened into a readable format for reporting.

	location = "Chicago"				location = "New York"				location = "Toronto"				location = "Vancouver"			
	item				*item*				*item*				*item*			
time	home	comp.	phone	sec.	home	comp.	phone	sec.	home	comp.	phone	sec.	home	comp.	phone	sec.
Q1	854	882	89	623	1087	968	38	872	818	746	43	591	605	825	14	400
Q2	943	890	64	698	1130	1024	41	925	894	769	52	682	680	952	31	512
Q3	1032	924	59	789	1034	1048	45	1002	940	795	58	728	812	1023	30	501
Q4	1129	992	63	870	1142	1091	54	984	978	864	59	784	927	1038	38	580

3 Key OLAP Operations

- **Roll-up:** Perform aggregation on a data cube by climbing up a concept hierarchy (e.g., from *City* to *Country*).
- **Drill-down:** The reverse of roll-up; moves from high-level summary to lower-level details (e.g., from *Year* to *Month*).
- **Slice:** Selects one particular dimension from a given cube and provides a new sub-cube.

- **Dice:** Selects two or more dimensions from a given cube and provides a new sub-cube.
- **Pivot (Rotate):** Rotates the data axes in view to provide an alternative presentation of data.

4 OLAP Architectures

Architectural Comparison

MOLAP	**Multidimensional OLAP:** Stores data in optimized multidimensional arrays. Fastest performance, but limited scalability.
ROLAP	**Relational OLAP:** Works directly with relational databases using SQL. Highly scalable, but query performance can be slower.
HOLAP	**Hybrid OLAP:** Combines MOLAP for aggregations (speed) and ROLAP for detailed data (scalability).

OLAP transitions data into "information" by allowing users to slice and dice through complexity.